EFY TESTED PREETI BHAKTA

HOME AUTOMATION SYSTEM
For Up To Four Devices

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This project is designed to control up to four home electrical devices using Node-RED, Raspberry Pi and local Wi-Fi network. Node-RED is a flow-based development tool for visual programming.

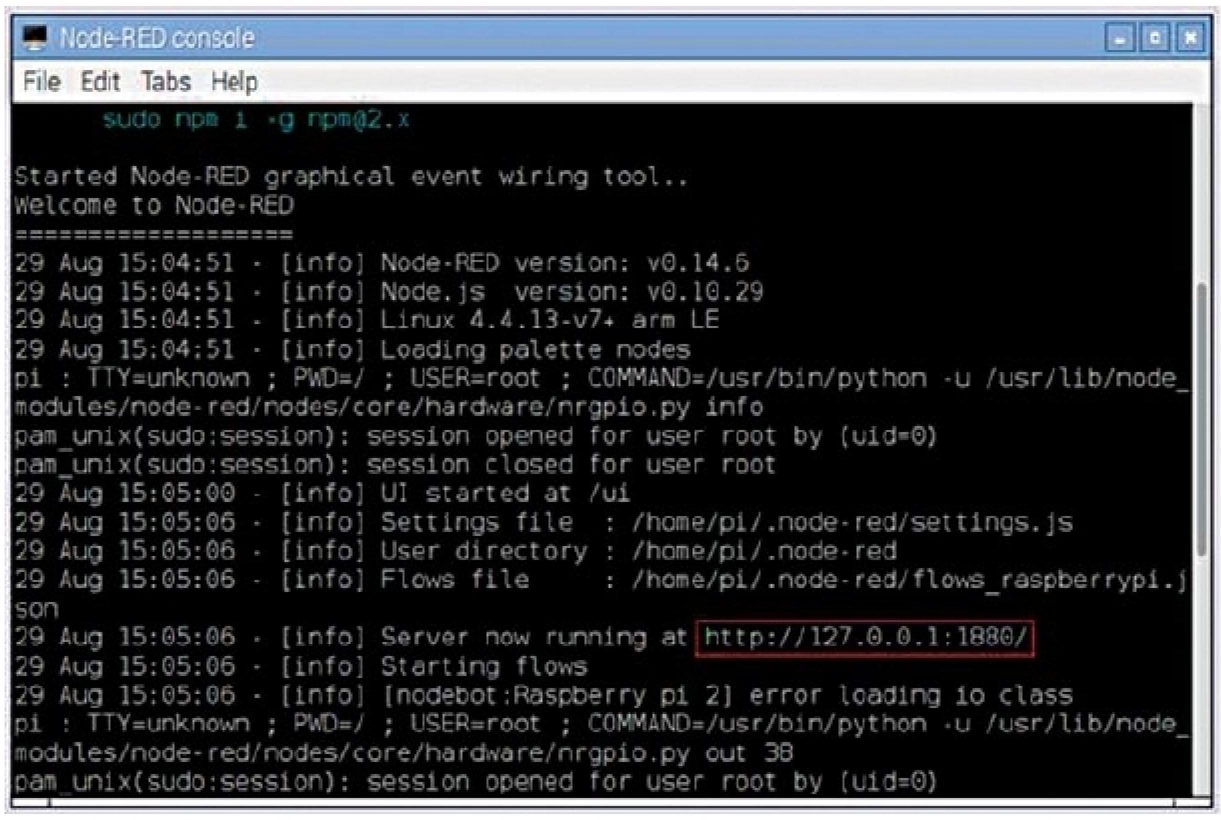

Fig. 1: Node-RED console

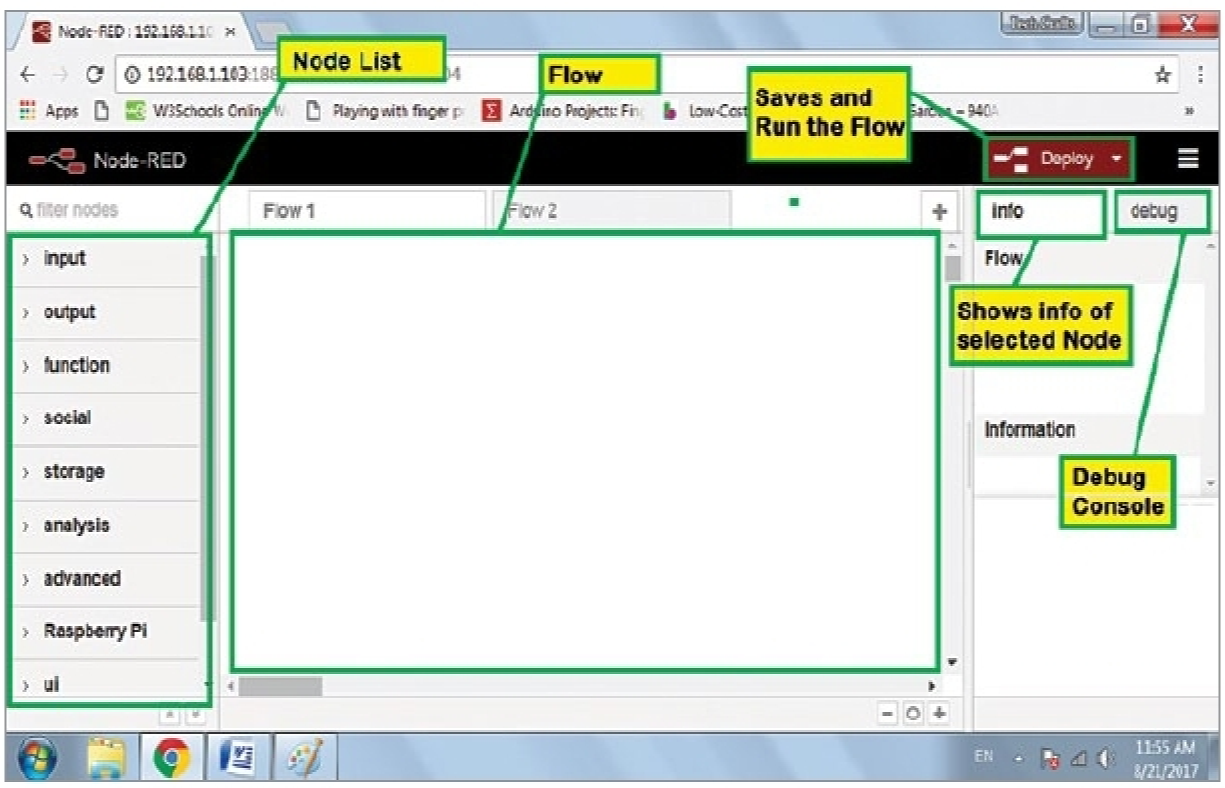

Fig. 2: Homepage of Node-RED

Node-RED

Node-RED is a graphical user interface (GUI)-based programming platform. It is a powerful platform to make Internet of Things (IoT) applications just by connecting its predefined code blocks like input, output, social media, Raspberry Pi GPIO and so on. You just need to connect predefined code blocks, called nodes, to perform a task. Connected nodes are collectively called flow.

Node-RED was developed as an open source project at IBM in 2013 to connect their hardware with Web-based applications, email servers, social media, etc. It has now become a popular IoT programming tool. When combined with Raspberry Pi, Node-RED becomes even more efficient for interacting with the physical world.

Preparing Raspberry Pi for Node-RED

The latest Raspbian operating system (OS) for Raspberry Pi comes with Node-RED application by default. If you want to install the latest version of Node-RED, run the following commands on the terminal:

```
sudo apt-get update
sudo apt-get upgrade
```

This will install the latest Node-RED on Raspberry Pi. If you are facing problems, install it manually by entering the following script on the terminal:

```
bash <(curl -sL https://raw.
githubusercontent.com/node-red/
raspbian-deb-package/master/resources/
update-nodejs-and-nodered)
```

Note that, after using this script to upgrade Node-RED, you will not be able to upgrade it again using apt-get upgrade command.

To open Node-RED from the desktop of Raspberry Pi, click on Raspberry Pi icon and then from the drop-down menu select Programming→Node-RED. Alternatively, you can type the following command on the terminal: